

Foundation Models for Edge Intelligence in Autonomous Driving: Challenges and Opportunities

Komala Subramanyam Cherukuri, Yunhe Feng, and Kewei Sha
University of North Texas, USA

komalasubramanyamcherukuri@my.unt.edu, {yunhe.feng, kewei.sha}@unt.edu

Abstract—Foundation models are emerging as an important direction for autonomous driving, bringing multimodal understanding, contextual reasoning, and knowledge informed decision-making. However, practical deployment requires foundation model workloads to meet the real-time, resource, and safety constraints of the vehicle-edge. In this position paper, we develop a vehicle-edge systems perspective on foundation model integration, examining the interactions among real-time operating systems, middleware, storage, heterogeneous computing resources, and vehicle autonomy. We further address foundation model execution and reasoning challenges under operational edge constraints, emphasizing multimodal input representation, data quality, communication, safety, security, and privacy. We also identify key research opportunities for enabling foundation model based edge intelligence in autonomous driving.

Index Terms—Foundation Models, Autonomous Driving, Edge Computing

I. INTRODUCTION

Modern mobility has been shaped by the advances in autonomous driving (AD) over the past decade. Autonomous vehicles operate across heterogeneous road environments, traffic behaviors, sensing conditions, and road-user interactions while meeting strict real-time and safety requirements within limited onboard computing resources. Current AD systems largely rely on conventional modular architectures, with perception, localization, prediction, planning, and control implemented as coordinated components [1], [2]. Despite significant progress, reliable operation remains challenging in rare, ambiguous, long-tail, and safety-critical scenarios, limiting broader real-world deployment [3], [4], [5].

Foundation models offer a promising direction for advancing autonomous driving by extending contextual reasoning and multimodal understanding across visual, linguistic, and other driving information, along with richer semantic scene understanding and spatial and temporal reasoning [6], [7], [8]. Large-scale pretraining also creates opportunities to improve generalization across unfamiliar driving environments and support adaptation to changing traffic rules, regional conditions, and driving contexts through language based knowledge [9], [10]. Foundation models can support knowledge informed high level decision-making and planning in complex driving situations while providing more interpretable explanations of driving decisions [11], [12].

Autonomous driving is inherently sequential, interactive, and safety-critical [13]. Foundation model reasoning must therefore remain grounded in current multimodal observations and vehicle state, produce decisions within required time limits, and generate outputs that can be validated before influ-

encing planning, control, or other safety-critical vehicle functions. On vehicle-edge platforms, foundation-model execution is constrained by available compute, memory, storage, and power resources while simultaneously handling heterogeneous multimodal inputs and varying reasoning workloads. Competition for shared resources, variability in inference latency, and changes in the freshness of model inputs can subsequently influence the timeliness and validity of generated outputs for real-time driving.

As shown in Table I, existing surveys remain largely model-centric, with limited treatment of foundation model execution and reasoning under vehicle-edge system constraints. To address this gap, this position paper establishes a *vehicle-edge systems perspective* on foundation model based autonomous driving. We examine the capacity of onboard vehicle-edge stack required to support foundation model intelligence, including real-time operating systems, middleware, storage, and heterogeneous computing resources. We further examine foundation model execution and reasoning challenges under operational edge constraints, including data quality, communication, security and privacy. Based on these considerations, we identify key research directions for practical foundation model integration for autonomous driving.

II. CHALLENGES AND OPPORTUNITIES

Foundation model integration for autonomous driving introduces challenges that extend from onboard system support to edge-aware model execution as shown in Figure 1. The following discussion examines these challenges and corresponding research opportunities from a vehicle-edge systems perspective.

A. On-board System Architecture

1) Real-Time Operating Systems

Real-Time Operating Systems (RTOS) provide the execution layer required to coordinate safety-critical and time constrained vehicle workloads. Automotive operating systems such as QNX, VxWorks [2], [25], and NVIDIA Drive OS [26] have supported conventional modular vehicle computing stacks by enforcing scheduling, isolation, and coordination across real-time and safety-critical tasks. However, deploying foundation models for autonomous driving introduces a *mixed-criticality co-execution challenge* by affecting the timing and dependability of co-existing safety-critical and operational workloads. The key challenge is therefore to preserve predictable system-level execution across mixed-criticality work-

TABLE I: Comparison with Existing Surveys on Foundation Models for Autonomous Driving

Reference	Context Reasoning	Resource Constraints	Safety-Aware Operation	Data Quality	Vehicle-Edge Stack	Summary
Cui <i>et al.</i> (2024) [8]	●	⦿	⦿	●	●	Surveys MLLM capabilities, driving applications, datasets, and future challenges.
Li <i>et al.</i> (2024) [14]	●	⦿	●	●	●	Surveys LLM-enabled human-like decision-making for AD.
Zhou <i>et al.</i> (2024) [15]	●	⦿	●	●	●	Surveys VLM tasks, applications, methods, metrics, and datasets in AD.
Gao <i>et al.</i> (2025) [16]	●	⦿	●	●	●	Surveys foundation models for planning, perception, reasoning, and simulation.
Jiang <i>et al.</i> (2025) [10]	●	⦿	●	●	●	Surveys VLA architectures, reasoning, safety, datasets, training, and evaluation.
Sathyam <i>et al.</i> (2025) [17]	⦿	●	⦿	●	●	Surveys foundation-model perception: knowledge, spatial, multisensor, temporal.
Sharshar <i>et al.</i> (2025) [18]	⦿	●	●	●	⦿	Surveys lightweight VLMs, compression, fine-tuning, and edge optimization.
Jing <i>et al.</i> (2026) [11]	●	⦿	●	⦿	●	Surveys MLLM perception, reasoning, datasets, architectures, and challenges.
Fourati <i>et al.</i> (2026) [19]	●	●	⦿	●	⦿	Reviews XLMs, ADS tasks, datasets, benchmarks, and deployment guidance.
Wu <i>et al.</i> (2026) [20]	⦿	●	⦿	●	●	Roadmaps FM development, AD applications, reliability, and deployment.
Cui <i>et al.</i> (2026) [21]	●	●	⦿	●	●	Surveys LLM for AD methods, pipelines, evaluation, and deployment.
Gao <i>et al.</i> (2026) [22]	●	⦿	●	●	●	Surveys foundation models for scenario generation and analysis in AD.
Tian <i>et al.</i> (2026) [23]	●	●	⦿	●	●	Surveys VLM integration in AVs: modular, end-to-end, evaluation.
Zhao <i>et al.</i> (2026) [24]	⦿	●	●	●	●	Surveys VLM human-vehicle interaction, pedestrian intent, and perception.
OURS	●	●	●	●	●	We establish a vehicle-edge systems perspective connecting foundation model reasoning, onboard systems, resource-aware execution, safety, and data quality.

Legend: ● = Yes; ⦿ = Partially, ● = No

loads under shared-resource contention on heterogeneous vehicle edge platforms. Addressing this challenge requires RTOS level execution control mechanisms that combine scheduling, isolation, and resource governance to preserve timing behavior and dependability across the vehicle computing stack.

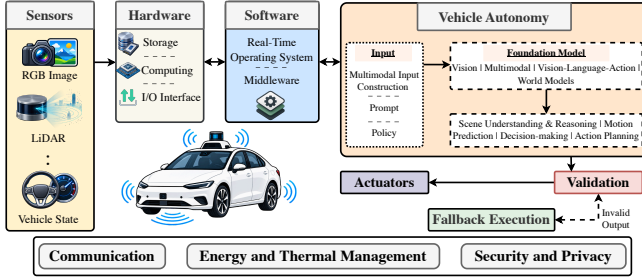


Fig. 1: Foundation Model Edge Intelligence Architecture for Autonomous Driving

2) Middleware

Existing middleware frameworks including ROS 2, DDS based communication frameworks [27], and Autoware support publish-subscribe communication, sensor data synchronization, quality of service (QoS) configuration, and timing-aware data exchange [28], [29]. However, foundation model based autonomous driving requires middleware to coordinate with downstream driving functions to ensure that generated outputs preserve temporal consistency, meet deadline constraints, and maintain message validity before they can support vehicle operation. This coordination is challenging because foundation model inference introduces substantial and variable computational demands, with response time and output structure changing with input complexity, driving context, and reasoning or generation requirements, making it difficult for middleware to maintain timely and predictable delivery with latency-sensitive vehicle functions [10], [15], [30]. As a result, a model output may be *computationally complete but operationally unusable* if it is produced from temporally inconsistent inputs, delivered beyond its validity window, or transmitted in a way that interferes with higher priority vehicle communication. Future

middleware for foundation model based autonomous driving should therefore dynamically manage the timing, freshness, and validity of model outputs through deadline monitoring, freshness control, priority-aware routing, bandwidth management, structured message handling, and fallback activation when outputs are late, incomplete, or no longer valid for the current driving context.

3) Storage

In autonomous driving systems, storage supports software artifacts, model files, runtime data, logs, updates, and rollback records [31]. Foundation model deployment at the vehicle-edge increases the demands placed on storage systems. For example, Mistral-7B occupies approximately 29.2–32 GB in full precision, whereas its quantized representation requires about 4.2 GB for edge-based reasoning in autonomous driving [32]. Even after compression, foundation model artifacts can remain gigabyte scale, which affects model download, staging, validation, update management, and rollback storage. A key opportunity is to develop *storage-aware model management* strategies that coordinate active model weights, quantized variants, safety configurations, update packages, and validated rollback versions while preserving access to the active model during vehicle operation.

Foundation model reasoning introduces additional storage requirements because vehicle decisions involve prompt context, sensor input, generated reasoning, action messages, timing metadata, model identifiers, and fallback outcomes [17], [23], [33]. A key opportunity is to define *selective logging policies* that determine which traces should be retained, compressed, summarized, or discarded under limited onboard storage. Such policies should preserve safety-critical, delayed, or invalidated model outputs for debugging, validation, and accountability while avoiding unnecessary retention of routine inference traces.

4) Vehicle Computing Resources

Modern autonomous vehicles rely on heterogeneous system on chip (SoC) platforms that integrate CPUs, GPUs, AI

accelerators, image signal processors, optical flow accelerators, memory controllers, sensor interfaces, and high-speed interconnects [1], [3]. Foundation model based autonomy builds on the same substrate while adding larger multimodal inference workloads requiring substantial compute, memory, and data movement across processing units. Evaluating vehicle computing resources, therefore, requires attention to peak acceleration capability, processor heterogeneity, memory subsystem efficiency, and data movement during model execution.

Vehicle-edge platforms are heterogeneous, and foundation model resource demands vary in parameter scale, modality composition, input length, and workload [34], [35]. Therefore, vehicle computing resource management for foundation model based autonomy should account for compute demand, memory occupancy, accelerator usage, and data movement under constrained edge resources while preserving resource availability for co-existing safety-critical functions.

Edge LLM and multimodal profiling studies show that accelerator offloading can redistribute resource use across processors [36]. OOM-Free Alpamayo [37] shows that Alpamayo-R1-10B requires 21.52GB of GPU VRAM, or about 23.7GB total runtime footprint, exceeding the 12–16 GB VRAM available on VRAM constrained discrete GPUs. To address this memory bottleneck, the paper uses software managed CPU–GPU swapping, layer-wise offloading, pipelined transfer compute overlap, and selective GPU-resident layers, reducing peak VRAM from 14.45GB to 3.99GB and inference latency from 14.52s to 4.09s without quantization or pruning. Nevertheless, the optimized model still imposes substantial memory and latency demands, suggesting that future vehicle computing optimization techniques must move beyond isolated foundation model execution and study how such workloads can be co-scheduled with other safety-critical tasks on the same onboard platform.

In contrast to discrete CPU-GPU platforms with separate system memory and GPU VRAM, vehicle-edge SoCs use unified memory architectures to reduce data movement and support low-latency execution [38], [39]. However, unified memory also brings foundation model tensors, runtime data, and co-executing vehicle functions into direct competition for the same memory capacity and bandwidth, even when their computation is distributed across CPUs, GPUs, and AI accelerators. Addressing this contention requires *memory-aware resource management* that coordinates accelerator placement, memory allocation, bandwidth use, and workload isolation under vehicle-edge constraints.

On vehicle edge platforms, sustained foundation model inference can increase heat generation and necessitate frequency reduction or workload throttling to keep hardware within thermal limits [3], [40]. Its energy demand also draws from the onboard power budget shared with sensing, actuation, communication, and safety functions, which can affect vehicle mobility. A key research opportunity is to develop energy and thermal aware resource management that adapts foundation model execution to onboard compute, power, and thermal conditions while preserving safety-critical vehicle functions.

5) *Vehicle Autonomy*

Conventional autonomous driving architectures have enabled substantial progress, but broader deployment remains limited by insufficient generalization, reasoning, and adaptation in rare, ambiguous, and safety-critical long-tail events. Foundation models can support decision-making in complex, ambiguous, or safety-critical driving scenarios [41], [42], [43]. However, full end-to-end integration across the driving stack remains an emerging research direction. Therefore, hybrid integration has emerged as a promising research direction, in which foundation models could be invoked selectively for decision-making and planning assistance in complex driving situations, while conventional perception, localization, control, and safety-monitoring modules continue to support real-time sensing, state estimation, actuation, constraint enforcement, and fallback execution [44]. *Hybrid integration* opens an important research opportunity for incorporating foundation model reasoning into conventional autonomous driving architectures under real-time constraints. Addressing this opportunity requires mechanisms for scheduling model outputs, managing compute and communication resources, validating relevance and freshness, converting outputs into bounded representations, and coordinating them with latency-sensitive vehicle functions.

B. *Edge-Aware Foundation Models for Real-Time Driving*

1) *Scenario-Level Dataset Construction*

For foundation model based autonomous driving, dataset design should move beyond isolated annotations toward linked representations of observations, reasoning, and actions. Recent datasets such as CARScenes [45], CoVLA [46], and OmniDrive [47] have moved in this direction by incorporating driving context, reasoning, and action-oriented supervision [22].

Despite recent progress, spatial grounding remains challenging for AD foundation models because reasoning and action prediction depend on metric cues such as depth, motion, and scene geometry. Recent large-scale efforts such as Alpamayo incorporate multi-camera driving observations and ego-motion with reasoning and trajectory supervision [34]. However, dataset construction for end-to-end foundation models must also preserve linkage across heterogeneous driving signals, language annotations, future trajectories, and ground-truth intermediate outputs. An open opportunity is to curate *unified driving scenario records* in which multi-view images, LiDAR, radar, maps, ego state, surrounding agent states, language annotations, future trajectories, and ground truth intermediate outputs are linked within the same driving event. A unified scenario structure would support modality and representation level trade-off analysis, robustness evaluation under deployment relevant conditions, and failure localization across perception, spatial grounding, motion prediction, and planning-level reasoning. Extending such datasets to long-tail events, adverse weather, and domain-specific driving conditions represents an important research direction for evaluating the robustness and generalization of foundation models beyond conventional urban benchmarks.

2) *Multimodal Input*

Foundation models process camera frames by encoding them into visual tokens and aligning them with the language model embedding space [48], [49], but they provide limited metric grounding; depth, scale, occlusion, free space, and inter-agent geometry remain difficult to preserve from camera observations alone. To introduce spatial structure, recent studies incorporate LiDAR and depth cues as calibrated 3D representations, providing stronger 3D grounding, drivable space, and agent relations [50], [51], [52]. Map and BEV representations provide structured road and top-down scene context [53], [54].

Driving data streams are heterogeneous, asynchronous, and multi-rate, varying in temporal sampling, spatial reference frame, and representation format [1], [55]. However, foundation model based driving must translate sensor, map, and state derived signals into a model compatible input representation while maintaining spatial and temporal alignment throughout input construction and across model input updates. A key research opportunity is *edge-aware representation construction* that preserves cross-modal and temporal input consistency for safety-critical driving decisions and supports efficient execution on vehicle edge hardware.

Adverse weather, poor visibility, occlusion, sensing limitations, and complex traffic interactions can degrade individual modalities or make a fixed set of sensor inputs insufficient, weakening the multimodal grounding required for foundation model reasoning and decision-making [56], [57]. Therefore, the trade-off between sensing reliability and processing cost motivates context-aware input policies that adapt the sensor modalities and representations provided to the foundation model according to driving conditions and task requirements.

3) *Edge-Efficient Model Design*

The computational costs of foundation models motivate model optimization for vehicle-edge platforms with strict memory, compute, and latency budgets. Among existing approaches, quantization improves memory and storage efficiency, while latency gains remain hardware and implementation dependent [58]. Pruning reduces parameters or processed tokens, but its effectiveness depends on model component sensitivity and preserving foreground and spatial information [59], [60]. A key research opportunity is to evaluate pruning trade-offs between computational savings and perception, reasoning, and planning performance across model components and domain-specific AD tasks.

Knowledge distillation reduces model scale by transferring knowledge from larger teachers to smaller students, but preserving driving-relevant information and meeting vehicle-edge resource constraints remain key considerations [61], [62]. This motivates the design of compact foundation models suitable for resource constrained vehicle-edge deployment. Accordingly, micro foundation models (μ FM), explored in IoT/CPS environments, provide a compact, domain and modality specific design that learns task agnostic representations through self-supervised pretraining [63], [64]. Extending these design principles to autonomous driving opens up two re-

search opportunities for vehicle-edge intelligence. First, micro foundation models present an opportunity to design compact models for nominal driving conditions with bounded sensing and reasoning demands. Second, hybrid autonomous-driving architectures (conventional + foundation models) motivate systematic investigation of micro foundation models as resource-efficient components for difficult or long-tail scenarios while limiting additional onboard resource demands.

Compression reduces the resource footprint of foundation models [65], yet vehicle-edge inference can remain constrained by accelerator memory and data transfer overhead. Therefore, efficient execution also depends on how model components are placed and transferred across available computing resources. An important research problem is to develop hardware-aware execution strategies that adapt model placement and offloading to resource availability while meeting latency and real-time driving requirements.

4) *Prompting, Policy Reasoning, and Action Planning*

Efficient edge execution must account for prompting, policy reasoning, and action planning. Recent approaches have introduced predefined reasoning stages, adaptive reasoning across different scenario complexities, and structured driving commands before trajectory generation [33], [66], [67], [68], [69], [70]. However, most studies emphasize reasoning and planning capability, with comparatively limited attention to the variability introduced across the prompt–reasoning–policy–action pipeline. For vehicle-edge deployment, such variability becomes a systems concern because changes in reasoning or policy can affect execution consistency, computational demand, and the resulting driving action.

For deployment in the foreseeable future, a practical direction is to reduce open-ended foundation model reasoning by predefining substantial parts of the reasoning and policy process while retaining limited flexibility for conditions that require additional contextual interpretation. Under nominal driving, for example, reasoning can follow an established sequence over factors such as traffic signals, lane availability, surrounding vehicles, and route intent, followed by selection from a defined set of maneuvers. Adverse conditions can require additional reasoning about visibility, road conditions, and sensor confidence, while increasingly complex or long-tail scenarios demand richer contextual reasoning and a broader reasoning process.

An open research problem is the design of these *partially predefined reasoning structures* for deployment-oriented autonomous driving. The key questions concern the reasoning steps that can be fixed for well understood operating conditions, while foundation model flexibility is introduced only at specific reasoning stages that require additional contextual interpretation. A second open problem is ensuring that the generated reasoning remains aligned with the resulting driving decision, particularly as more open-ended reasoning is introduced for complex scenarios.

For vehicle-edge systems, reasoning efficiency depends on update frequency, computational cost, and latency allocated to each reasoning update. Recent approaches such as AutoVLA

already vary reasoning depth across driving conditions [70]. A key direction is to adapt the *policy-relevant reasoning workload*, limiting computation to the context needed to support the current driving policy while allowing additional reasoning as ambiguity and decision complexity increase.

5) *Safety-Aware Edge Intelligence*

Autonomous driving operates under real-time constraints. Prior work shows that, at urban speeds of approximately 40 km/h, end-to-end response latency should remain below 100 ms to support safety-critical vehicle control [71], [72]. Foundation models introduce contextual reasoning into the driving decision process. However, ambiguous or complex scenarios may require additional reasoning before a decision can be made. When additional reasoning delays the current decision, the driving environment may change before the decision is produced, reducing its temporal validity and the system’s responsiveness to new events. Therefore, reasoning quality and decision timeliness can conflict when complex driving scenarios require prolonged reasoning before a decision can be reached. This motivates the design of *scenario-dependent reasoning budgets* that bound reasoning time or depth according to decision urgency, ensuring that additional reasoning does not extend beyond the available decision time.

Foundation models can produce hallucinated or insufficiently grounded reasoning that is inconsistent with the observed driving scene, introducing additional safety risks for decision making [73], [74]. A practical deployment path is to place foundation models within predefined, verifiable decision structures and explicit safety constraints, allowing generated decisions to be checked before execution. However, as autonomous driving moves toward end-to-end foundation model decision making, such external safeguards may no longer exist as separate components. *Runtime validation* therefore becomes increasingly important to verify generated decisions against current sensor observations, vehicle state, and safety constraints before they affect vehicle behavior.

Conventional autonomous driving safety requirements, testing approaches, and driving metrics remain applicable to foundation model based systems. However, foundation models also introduce language based reasoning, semantic decisions, and prompt conditioned behavior alongside driving outputs. Recent studies have begun to evaluate these systems through full-stack testing, semantic reasoning metrics, driving performance, and system-level measurements, but these dimensions remain assessed using different criteria and metrics [75], [76], [77]. A key direction is therefore to develop a *unified system-oriented evaluation methodology* that connects reasoning and decision quality with driving performance, safety, latency, and vehicle-edge resources.

Autonomous driving scenes evolve continuously, but route, lane, traffic rules, and relationships with tracked road users often remain valid across consecutive observations. Existing edge-LLM studies show that environmental and agent information can remain stable across multiple observations, while motion information often requires more frequent updates as the driving scene changes [75]. Therefore, reconstructing the same

context and reasoning at every update can impose additional vehicle-edge computing cost, while retaining outdated reasoning after meaningful scene changes can reduce the reliability of the resulting driving decision. A promising direction is therefore to maintain a *persistent reasoning state*, preserving valid reasoning and updating only the components affected by meaningful scene changes, with reasoning refreshed at a frequency consistent with downstream planning requirements.

C. *Data Quality for Foundation Models at the Vehicle-Edge*

Data quality is a key requirement for foundation model enabled edge intelligence in autonomous driving. Existing AD architectures address several data quality requirements through real-time operation, safety-critical validation, and system timing constraints. However, as summarized in Table II, redundancy, completeness, and consistency remain key trade-offs between data sufficiency, downstream task reliability, and vehicle edge resource use.

Foundation models offer a promising direction for balancing data quality and edge efficiency by enabling scene level reasoning over selected multimodal data while preserving completeness and consistency across sensor and intermediate representations [81], [65]. Human driving suggests that effective scene understanding can be achieved through selected viewpoints rather than uniform processing of all available viewpoints. In contrast, autonomous vehicles rely on dense multi-sensor coverage to support robustness, safety, and wide area perception, creating substantial overlap and edge resource pressure. This contrast motivates a future research direction in which foundation models are explored as *redundancy aware reasoning components* that identify informative multimodal evidence while preserving scene coverage and consistency for downstream driving functions. In this direction, foundation models could support the selective use of overlapping sensor information to reduce redundant processing while preserving the multi-sensor robustness. Redundancy aware use of sensor data must preserve completeness and consistency by retaining task required scene coverage and supporting a valid shared scene state. Edge efficiency improvements should therefore be evaluated against the ability of reduced representations to satisfy downstream autonomous driving task requirements.

Foundation models introduce additional quality requirements across the input, prompt, reasoning, and output stages. Input quality affects the driving context provided to the model, prompt quality shapes task interpretation, reasoning quality reflects the validity of the generated reasoning, and output quality determines the reliability of the resulting decision. A key direction is to develop quality frameworks, benchmarks, and evaluation methods that assess these stages and capture quality propagation across the reasoning pipeline.

Annotation quality in datasets for foundation models is an important consideration, as the same driving scene can support more than one reasonable interpretation, reasoning path, or action [82], [83]. The resulting ambiguity becomes challenging when sensor observations, language descriptions, reasoning, and actions must be linked consistently. Annotation quality can

TABLE II: System Oriented Data Quality Dimensions for Conventional Modular Autonomous Driving Architecture

Dimension	Architectural basis in conventional AD stacks	Vehicle Edge Consideration
Redundancy [78], [79]	Multiple cameras, LiDAR, and radar often observe the same objects and scene regions, and this overlap can be carried forward into perception outputs, tracking states, and prediction inputs.	Redundant observations can improve robustness, but processing all overlapping streams at the edge increases inference workload and resource pressure.
Completeness [78], [80]	Completeness requires selected data to preserve sufficient scene coverage, task-required inputs, and contextual information for the current driving function and subsequent learning.	Incomplete data can leave downstream modules with insufficient scene coverage, or model context, increasing the risk of invalid fusion and degraded task performance.
Consistency [78], [80]	Consistency is required across multi-sensor data, map context, annotations, timestamps, and intermediate outputs used throughout the perception to planning pipeline.	Inconsistent inputs can prevent construction of a valid shared scene state, leading to unreliable fusion, unstable tracking, conflicting predictions, or degraded planning.

be further affected by noisy, incomplete, or poorly grounded labels, which can distort these links and propagate errors into model reasoning and downstream decisions. Addressing this challenge requires development of scalable validation methods that account for ambiguity while preserving spatial, temporal, and observation–reasoning–action consistency.

D. V2X Communication

V2X communication is evolving from exchanging raw sensor data, intermediate features, detections, and predefined messages toward foundation model oriented semantic exchange, where cooperative information is represented as multimodal tokens, scene descriptions, contextual knowledge, or intent for downstream reasoning and planning. Recent studies have explored this shift through VLM based vehicle infrastructure fusion, LiDAR to token representations, and time-indexed language-based knowledge retrieval [84], [85], [86], [87]. For example, V2X-UniPool [85] reduces the V2X transmission payload from 1.24×10^7 bytes in V2X-VLM to 1.51×10^5 bytes, with an estimated transmission delay of approximately 12 ms at 100 Mbps. While such representations reduce communication volume and provide richer context, they introduce an edge-efficiency trade-off between communication cost, preprocessing and inference overhead, and timeliness. A key research direction is therefore *context and resource-aware V2X processing* that dynamically determines what information to exchange, in which representation and resolution, at what update rate, and where to process it according to available edge resources and current driving requirements.

A promising research direction is *V2X-assisted fallback* under degraded reasoning or resource constraints. When local foundation model inference becomes unreliable or cannot meet the required compute or latency budget, available vehicles or infrastructure with greater computational capacity could provide targeted reasoning support or fallback guidance, enabling the ego vehicle to transition safely toward a minimal-risk state.

E. Security & Privacy

Foundation models introduce new privacy and security concerns across the autonomous-driving pipeline, spanning training data, pretrained components, multimodal inputs, prompts, intermediate representations, generated outputs, and downstream actions. Existing studies have demonstrated data poisoning, backdoor attacks, compromised vision encoders, domain-generalized triggers, token-level manipulation, and action-level hijacking, while current defenses provide only partial protection [88], [89], [90], [91]. An important research

direction is to develop *lightweight edge-side protection* that verifies model integrity, multimodal inputs, intermediate representations, and generated actions without excessive computation, memory, or latency overhead. Preserving privacy is equally critical, particularly for protecting sensitive sensor data, prompts, contextual information, and exchanged model representations during local and V2X assisted processing.

III. RESEARCH AGENDA AND CONCLUSION

The research directions developed in this paper emphasize the need for foundation model based autonomous driving to become more predictable, controllable, and adaptable under vehicle-edge constraints. For well-defined nominal driving conditions, compact and task-focused foundation models can operate with more deterministic execution through predefined reasoning structures, constrained policies, and limited model flexibility. Establishing such controlled systems can provide a practical baseline for studying how model scale, reasoning depth, multimodal input demands, and computational cost should evolve as driving conditions become more adverse, ambiguous, or complex.

Foundation model design must be considered together with the vehicle computing stack, rather than treating model optimization as an isolated problem. Future research should integrate foundation model design with execution scheduling, middleware coordination, storage, shared computing resources, and safety-critical co-execution to translate model efficiency into predictable onboard operation. Evaluation should consequently move toward deployment-oriented baselines that assess reasoning and driving performance together with latency, compute and memory use, energy and thermal behavior, input validity, and execution consistency across heterogeneous vehicle-edge platforms. In conclusion, realizing the potential of foundation models for autonomous driving requires moving from model centric development toward efficient, predictable, and deployment-oriented edge intelligence. The research agenda presented in this paper outlines a path toward practical integration by connecting model design, vehicle-edge execution, and system-level evaluation.

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